SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO-2 : Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)

Assessment Tool : Short Answer Test

Weightage: 10%

QUESTIONS: **Total Marks: 20**

Question 1 (5 Marks)

Suppose that a company wants its website to work seamlessly on mobile, tablet, and desktop devices. Explain the concept of responsive web design and its importance. Describe the key CSS techniques used to achieve responsiveness, and explain how CSS Flexbox or Grid can be used to design adaptive layouts. Illustrate your answer with a simple example layout and suggest improvements to enhance user experience across different devices.

Question 2 (5 Marks)

Suppose that a webpage contains a form where user input must be validated before submission. Explain the concept of client-side validation and its purpose. Describe how JavaScript event handling is used in validation and explain the role of regular expressions in validating inputs such as email or phone numbers. Provide a suitable example and suggest improvements for better usability and security.

Question 3 (5 Marks)

Suppose that a webpage layout breaks when the browser window is resized. Explain the CSS box model and its components, and describe how CSS positioning techniques affect layout structure. Identify possible causes of layout breaking and explain how media queries can be used to resolve these issues. Suggest best practices to ensure a stable and responsive layout.

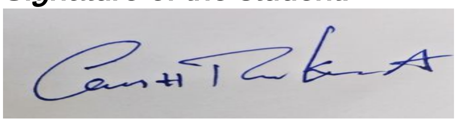
Question 4 (5 Marks)

Suppose that a website requires dynamic content updates without reloading the entire page. Explain the concept of the Document Object Model (DOM) and its role in web development. Describe how JavaScript can be used to manipulate the DOM for updating webpage content dynamically. Provide an example and suggest improvements to enhance performance and user experience.

Code of Conduct:

I **Abburi Ganesh Ram Kamal** *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding	Good understanding with	Basic understanding with some misconceptio	Poor or incorrect understandi

		of the concept	minor gaps	ns	ng
Accuracy of Answer	**25**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with	Incorrect or irrelevant answer
				noticeable errors	
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**10**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Assessment Tool-3 CSA4305 IP

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1A. Responsive web Design:

Responsive web Design (RWD) is a web design approach that allows a website to automatically adjust its layout and Content to fit different screen sizes such as mobile Phones, tablets and desktops

Importance:

- Provides a better user Experience on all devices
- Supports mobile, tablet and desktop screens
- Improves SEO and website accessibility.
- Reduce development and maintenance costs.
- uses a single website for all devices.

Key CSS Techniques:

1. Media Queries - Apply different style for different screen sizes.
2. Flexible Layouts - use %, em, rem, vw and vh instead of fixed pixels.
3. Flexible Images - use:

```
img {
    max-width: 100%;
    height: auto;
}
```

4. Responsive Typography - use relatives font sizes like rem and em.

Using CSS Flexbox

Flexbox creates Flexible layouts the automatically adjust to different screen sizes.

Example:
```
.Container {
    display: flex; }
@media (width: 768px) {
    .Container {
        Flex-direction: colum;
    }}
```

- Desktop: Menu | Content | Sidebar
- Mobile: Menu → Content → Sidebar (stacked vertically)

Improvements:

- use a responsive navigation menu
- optimize images For Faster loading
- make buttons touch-friendly
- use readable fonts and proper spacing.
- Test the website on different devices.

working:
- on desktop, the three sections appear side by side
- on mobile, they automatically stack vertically.

Conclusion:

Responsive web Design ensures websites works smoothly on all devices by using media queries, Flexible Layouts and Flexbox/ Grid.

2. Client-side validation:

Client-Side validation is the process of checking user input in the browser using Javascript before the Form is Submitted to the server. It Ensures that the entered data is Correct and Complete.

Purpose:

- Reduce invalid Form submissions.
- Improves user Experience by providing Instant Feedback
- Reduce Server Load
- Save time by detecting Errors before submission.

Javascript Event Handling in validation

Javascript uses event such as submit, input, change and blur to validate Form Fields.

Example:

```
< Form onsubmit = "return validateForm()">
    <input type = "email" id = "email">
    <input type = "submit" value = "submit">

 </Form>

<script>
Function validateForm () {
        Let email = document. getElementById("email").
              values;
        if (email == "") {
               alert(" Email is required");
```

return false;

} return true;

</script>

Role of Regular Expression (Regax):

Reguar Expression (Regax) are patterns used to validate input formats.

Email validation:

Let email pattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;

Phone number validation (10 digits):

Let phone Pattern = \^[0-9]{10}$ /;

Regex ensures that the input follows the Required Format.

Improvement:

- Display clear Error messages near input fields
- use HTML5 validation (required, type="email", pattern).
- validate data on the server as well for security
- Sanitize inputs to prevent attacks such as SQL injection and XSS.
- Highlight invalid fields for better user experience.

Conclusion:

Client-side validation uses JavaScript events and regular expressions to verify user input before form submission. It improves user experience and reduces sever load but server-side validation is also necessary to complete security.

3. CSS Box model and Responsive Layout

CSS Box mode:

The CSS Box model describes how element are displayed on a webpage. Every HTML element is treated as a rectangular box consisting of four parts:

1. Content - The actual text or image
2. Padding - space between the content and the border
3. Border - Surrounds the padding and content
4. Margin - space outside the border that separates elements

CSS Positioning Techniques:

CSS Positioning controls where elements appear on the page

- static - Default Position.
- Relative - Moves relative to its normal position.
- absolute - Positioned relative to the nearest positioned ancestor
- Fixed - stays fixed even when the page is scrolled.
- sticky - Acts like relative until scrolling then become fixed.

Causes of layout Breaking:

- using fixed width
- Large images without responsive sizing
- Incorrect use of absolute positioning.

- missing media queries.
- Long text or overflowing content.

Using media Queries:

media queries apply different CSS style based on screen sizes.

Example:

```
.container {
    display: flex;
}
@media (max-width: 768px) {
    .container {
        flex-direction: column;
}}
```

Result:

Desktop: Elements appear side by side

mobile: Elements stacks vertically, preventing layout breaking.

Best practices:

- use Flexbox or CSS Grid for flexible layouts
- use relative units (%, em, rem, vw, vh) instead of fixed pixels.
- makes image responsive: img { max-width: 100%; height: auto; }

Conclusion:

The CSS Box model and positioning techniques determine how webpages elements are designed. layouts may Break due to fixed sizes or improper positioning. using media queries, Flexbox / Grid and responsive CSS techniques ensures a stable flexible and user-friendly.

4A. DOM and Dynamic Content updates

Document object model (DOM):

The Document object model (DOM) is a programming Interface for HTML and XML documents. It represents a webpage as a tree structure of object allowing JavaScript to access and modify elements, attributes and content dynamically without reloading the page.

Role of DOM in web development:

- Allows JavaScript to access HTML element
- Dynamically updates webpage content
- changes styles and attributes
- Responds to user actions such as clicks and key presses.
- Creates interactive web applications.

Java script DOM manipulation:

Java script uses methods such as:

get Element By Id ()

query selector()

Inner HTML

text Content

Style.

Example:

<P Id="msg"> welcome! </P>

<button onClick="changeText()"> clickme</button>

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```
<Script>
function changeText() {
    document.getElementById("msg").InnerHTML=
        "Hello, User!";
}
</Script>
```

Output:

- Before clicking: welcome
- After clicking: Hello, User! (without reloading the page)

Improvements:

- use textContent instead of inner.HTML when adding plain text for better security.
- Minimize unnecessary DOM updates to improve performance
- use event listeners (addEventListener) instead of inline events.
- optimize large updates using DocumentFragment or Efficient DOM manipulation.
- validate user input to prevent security issues like XSS

Conclusion:

The DOM enables Javascript to dynamically access and modify webpages elements without reloading the page. By using DOM manipulation method developers can create fast, interactive and user-friendly web applications while improving performance and security.